

BUILD YOUR
IDEAS.
CRATE THE
FUTURE!

ONE-WEEK SUMMER Training Camp on Robotics



Program Overview

This one-week Summer Training Camp is designed around the principles of STEM Education (Science, Technology, Engineering, and Mathematics), using Robotics as a powerful learning tool. The program transforms students' classroom knowledge into exciting, hands-on experiences where concepts are understood through learning by doing rather than only reading.

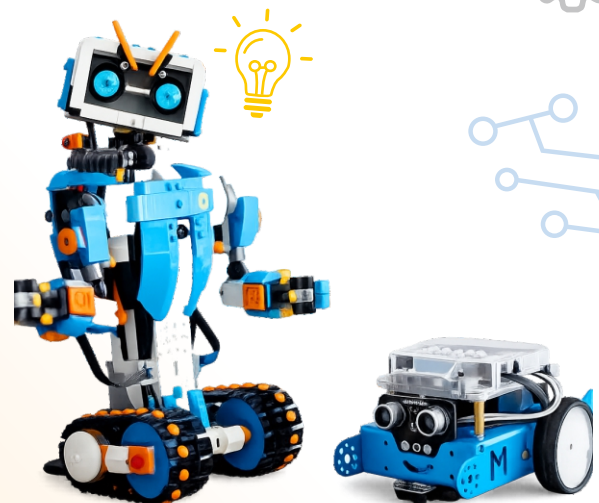
Through robotics-based activities, students apply scientific and mathematical ideas while building models, designing simple engineering systems, and solving real problems. This practical exposure helps children clearly understand how STEM concepts work together in real life.

The camp provides an inclusive platform where every learner—especially average and below-average students—can experience the joy of Science and Technology in a supportive environment. By engaging in creative robotics tasks and problem-solving challenges, students naturally develop critical thinking, logical reasoning, teamwork, and an engineering mindset.

Strongly rooted in STEM fundamentals, this program emphasizes creativity, experimentation, and hands-on learning through Robotics, enabling students to gain meaningful practical knowledge beyond textbooks and inspiring them to think, explore, and innovate like young engineers.

Duration:
1 Week (6 Days)

Batch Size:
60 Students



Learning Module

Day- 1

Fundamentals & Live Demonstrations

Introduction to Robotics

- ▀ Basics of Robotics and its importance in modern industries
- ▀ Demonstration of real robotic systems:
 - Humanoid Robot
 - 9 DOF Biped Robot
 - Spider Robot
 - Mini Robot
 - Droid Robot
 - Otto Bot
 - Robo Dog
 - Smart AI Robotics Car
 - Robotic Arm

Day- 2

Arduino & Coding

Arduino IDE & Basic Hardware Interfacing

- ▀ Introduction to Arduino IDE
- ▀ Writing and uploading programs
- ▀ Hands-on activities:
 - LED Blinking
 - Fading Effects
 - Strobe Light

Day- 3

Smart Signalling and Lighting Automation

- ▀ Smart Signalling and Lighting Automation introduces students to the basics of analog and digital signals through practical Arduino projects. By building a traffic light system and an automatic street light using sensors, learners understand how real-world automation works and how sensor inputs are converted into intelligent actions.
 - Traffic Light Development
 - Automatic street Light development

Learning Module

Day- 4

Sensor Core Robotics

- ▀ Understanding Robotics from the sensor level—how robots see, sense, decide, and act.
- ▀ A relative study between human sensors and Robots sensors.
 - Development of Laser Security System
 - Development of water Alarm

Day- 5

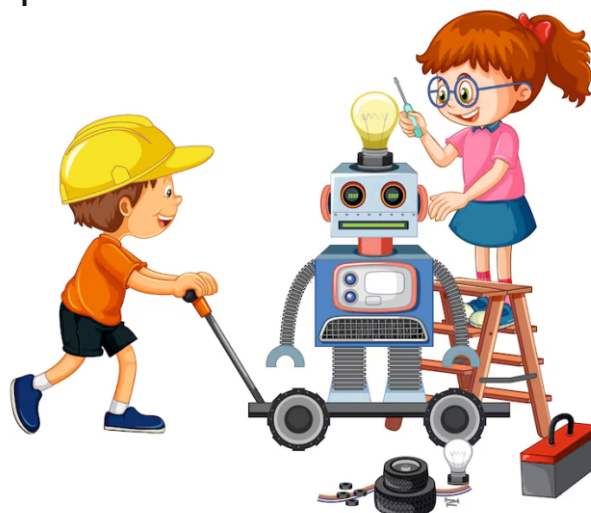
Module 9. Line Following Robot

- ▀ Working principle of IR sensors
- ▀ Robot navigation logic
- ▀ Hands-on development of:
 - Line Following Robot
 - Applications in industrial automation

Day- 6

Obstacle Avoiding Robot

- ▀ Introduction to ultrasonic (sonar) sensors
- ▀ Motor driver (L298N) working
- ▀ Hands-on development using:
 - Atmel Microcontroller
 - Ultrasonic Sensor
 - Motor Driver
- ▀ Autonomous navigation concepts



Learning Outcomes

By the end of the Summer Camp:

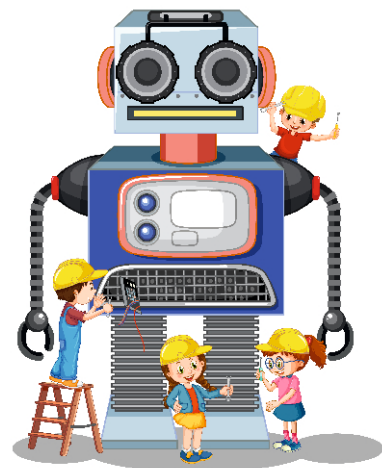
- ▀ Improves understanding of Science and Mathematics through practical Robotics activities.
- ▀ Encourages learning by doing and experiential learning
- ▀ Develops critical thinking and problem-solving skills
- ▀ Builds creativity, innovation, and logical reasoning
- ▀ Enhances confidence through hands-on project building
- ▀ Introduces students to STEM concepts in an engaging way
- ▀ Promotes teamwork, collaboration, and communication skills
- ▀ Helps average and below-average students grasp concepts easily through practice.
- ▀ Creates early interest in Robotics, AI, IoT, and emerging technologies
- ▀ Inspires students to think like engineers and innovators rather than passive learners

Certification

- ▀ Summer Training Camp Certificate

Who Can Join?

- ▀ Students from Standard V-XII.



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